



MODULUS OF COMPLEX NUMBERS



MODULUS OF COMPLEX NUMBERS

DEFINITION

For a complex number $z = x + iy$, the modulus (or absolute value) of z , denoted by $|z|$, is defined by

$$|z| = \sqrt{x^2 + y^2}$$

EXPLANATION

- The modulus of $z = x + iy$ represents the distance of the point $P(x, y)$ from the origin O in the Argand (complex) plane.
- It is the length of the line from the origin to the point $P(x, y)$.
- This line is the hypotenuse of a right triangle with sides of length x (along the real axis) and y (along the imaginary axis).
- The modulus is always non-negative, i.e., $|z| \geq 0$.

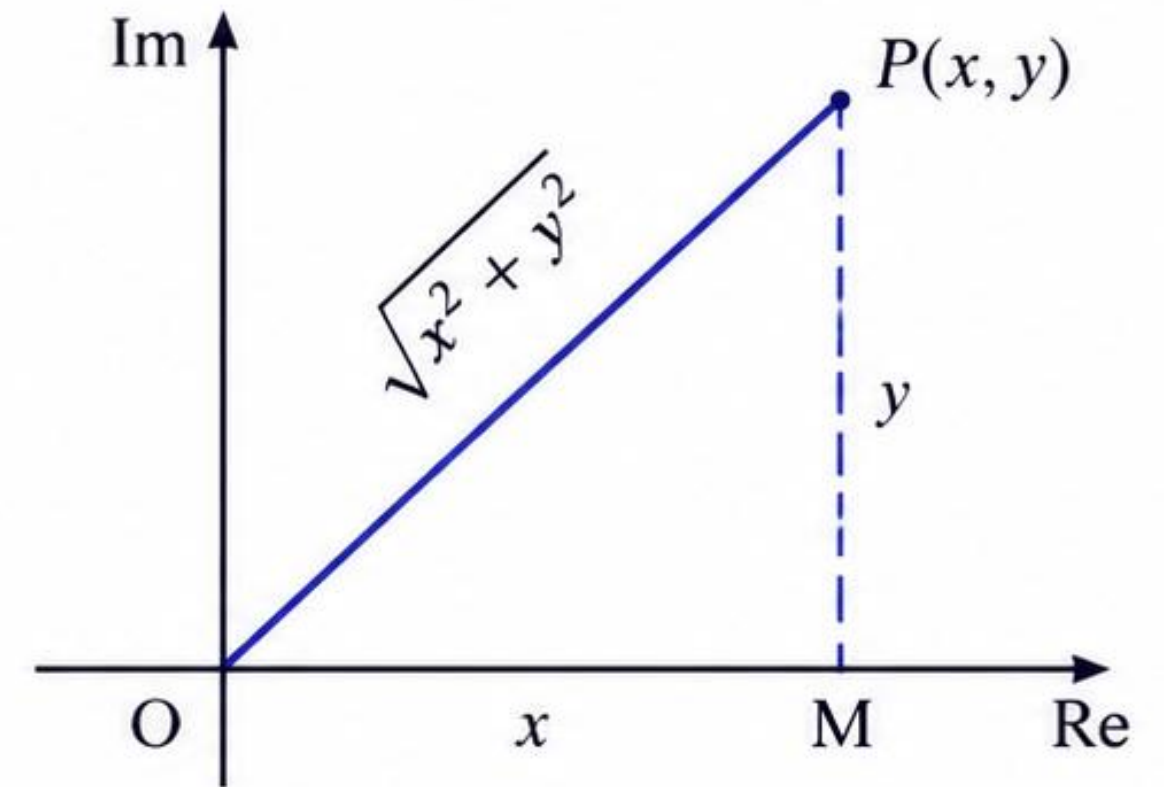


Fig. 2.16

Geometrical representation of $|z|$ in the Argand plane.



★ EXAMPLES OF MODULUS

For a complex number $z = x + iy$, $|z| = \sqrt{x^2 + y^2}$.

Example 1

Find $|i|$.

$$\begin{aligned}|i| &= \sqrt{0^2 + 1^2} \\ &= \sqrt{1} \\ &= \boxed{1}\end{aligned}$$

Example 2

Find $|-12i|$.

$$\begin{aligned}|-12i| &= \sqrt{0^2 + (-12)^2} \\ &= \sqrt{144} \\ &= \boxed{12}\end{aligned}$$

Example 3

Find $|12 - 5i|$.

$$\begin{aligned}|12 - 5i| &= \sqrt{12^2 + (-5)^2} \\ &= \sqrt{144 + 25} \\ &= \sqrt{169} \\ &= \boxed{13}\end{aligned}$$

IMPORTANT RESULT

For any complex number $z = x + iy$,

$$\boxed{|z|^2 = z\bar{z}}$$

Where $\bar{z} = x - iy$ is the conjugate of z .

That is,

$$(x + iy)(x - iy) = x^2 + y^2 = |z|^2$$





★ PROPERTIES OF MODULUS

For any complex numbers z_1, z_2 ,

1

$$|z| = |\bar{z}|$$

The modulus of a complex number is equal to the modulus of its conjugate.

2

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

Triangle inequality for complex numbers.

3

$$|z_1 z_2| = |z_1| |z_2|$$

The modulus of a product is equal to the product of the moduli.

4

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

The modulus of the difference is greater than or equal to the absolute difference of the moduli.



MORE PROPERTIES

For any complex numbers z_1, z_2 (and $z_2 \neq 0$ where required), the following properties hold:

1

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad z_2 \neq 0$$

The modulus of a quotient is equal to the quotient of the moduli.

2

$$|z^n| = |z|^n, \quad \text{where } n \in \mathbb{Z}$$

The modulus of a power is equal to the power of the modulus.

3

$$\operatorname{Re}(z) \leq |z|$$

The real part of a complex number is less than or equal to its modulus.

4

$$\operatorname{Im}(z) \leq |z|$$

The imaginary part of a complex number is less than or equal to its modulus.

5

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \quad z_2 \neq 0$$

The modulus of a quotient is equal to the quotient of the moduli.

6

$$|z^n| = |z|^n, \quad \text{where } n \in \mathbb{Z}$$

The modulus of a power is equal to the power of the modulus.

7

$$\operatorname{Re}(z) \leq |z|$$

The real part of a complex number is less than or equal to its modulus.

8

$$\operatorname{Im}(z) \leq |z|$$

The imaginary part of a complex number is less than or equal to its modulus.

TRIANGLE INEQUALITY

For any complex numbers z_1 and z_2 ,

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

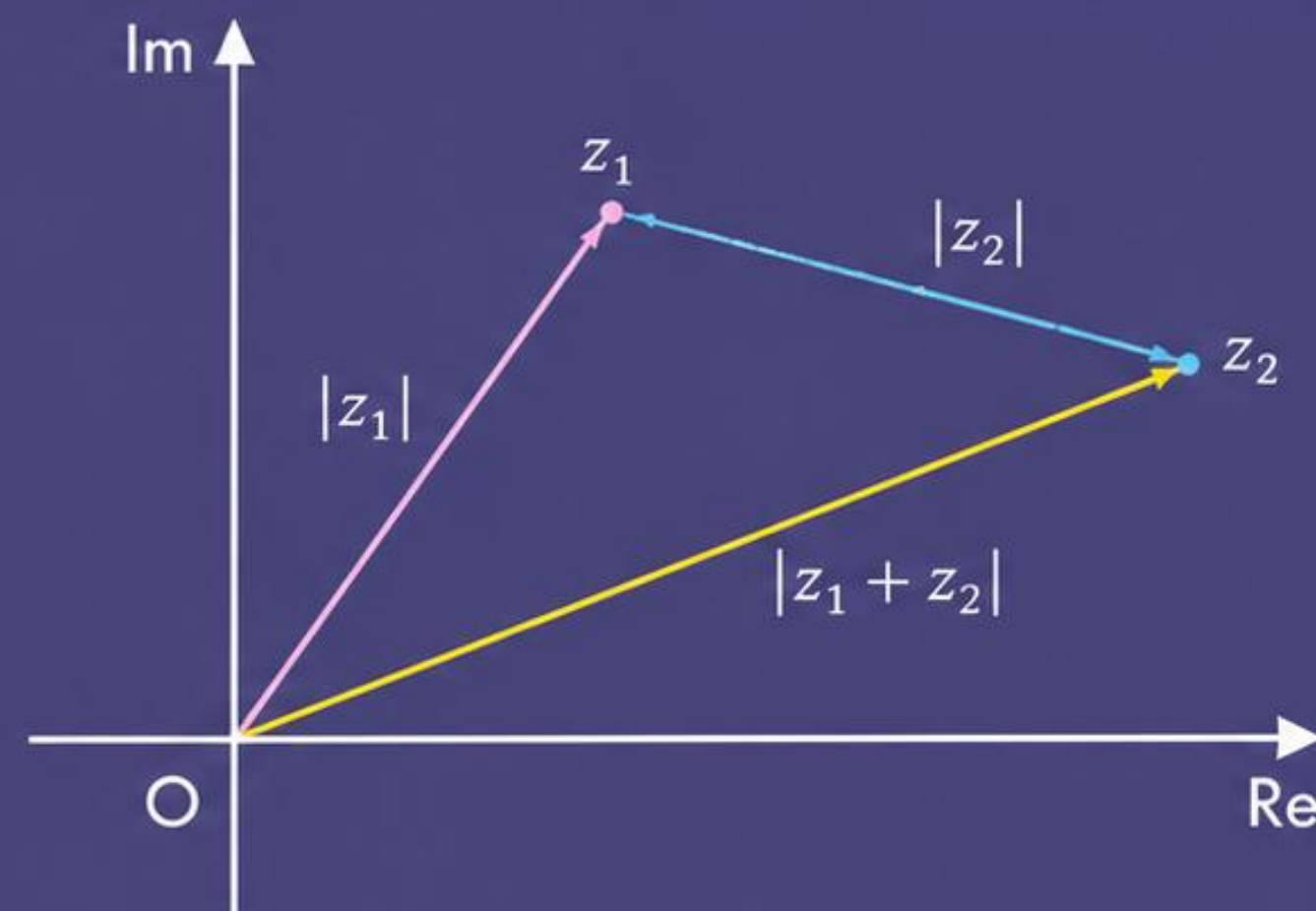
GEOMETRICAL INTERPRETATION

- Let z_1 and z_2 be represented by vectors $\vec{Oz_1}$ and $\vec{Oz_2}$ in the Argand plane.
- The sum $z_1 + z_2$ is represented by the vector $\vec{Oz_2}$.
- In the triangle Oz_1z_2 , the length of the side Oz_2 (i.e., $|z_1 + z_2|$) cannot be greater than the sum of the other two sides (i.e., $|z_1| + |z_2|$).
- Hence, $|z_1 + z_2| \leq |z_1| + |z_2|$.

GENERAL FORM

$$\begin{aligned} &|z_1 + z_2 + z_3 + \cdots + z_n| \\ &\leq |z_1| + |z_2| + |z_3| + \cdots + |z_n| \end{aligned}$$

for $n = 2, 3, \dots$



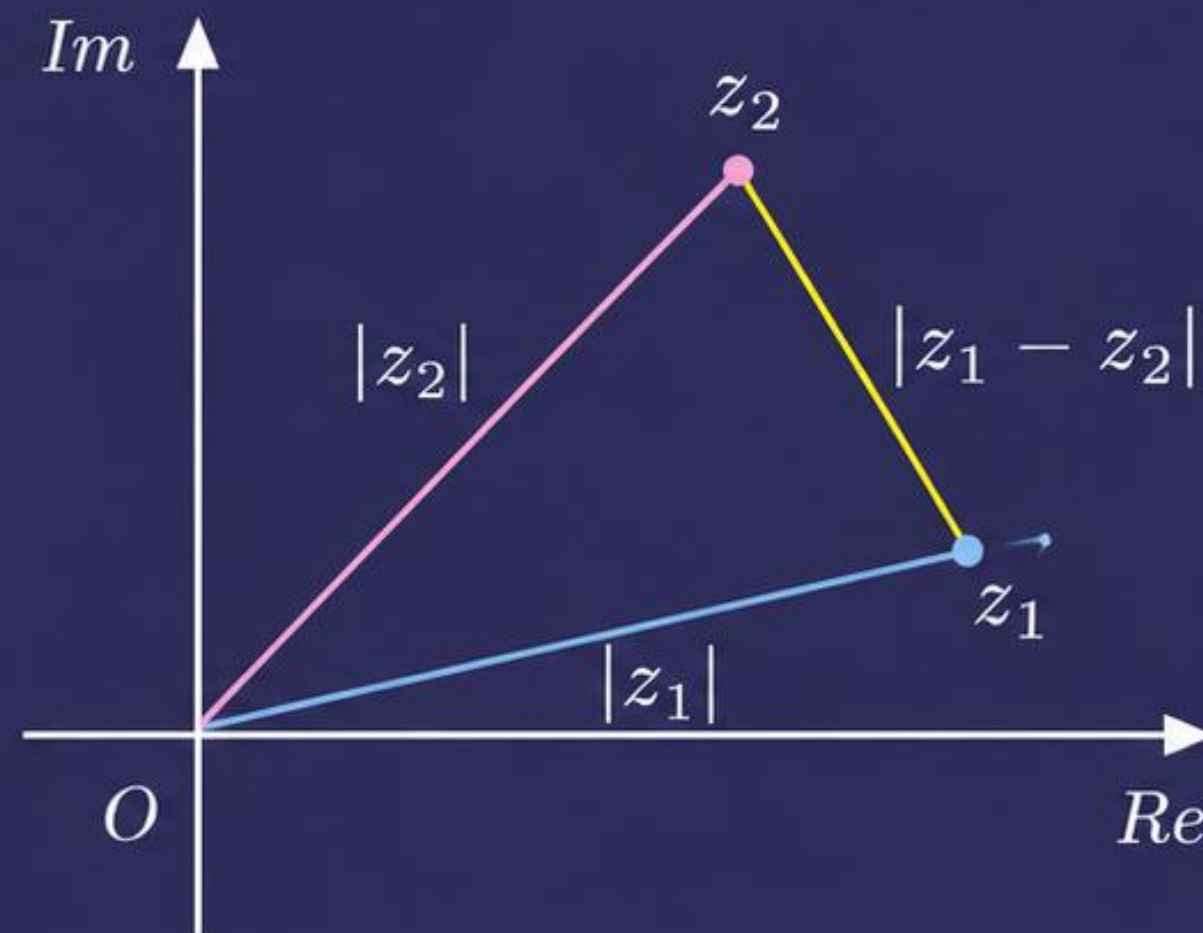
Since the length of the third side of a triangle is not greater than the sum of the other two sides, we get $|z_1 + z_2| \leq |z_1| + |z_2|$.

DISTANCE BETWEEN TWO COMPLEX NUMBERS

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then

$$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Hence, $|z_1 - z_2|$ gives the **distance between** the points z_1 and z_2 in the complex plane.



REMARK

For any complex numbers z_1 and z_2 ,

- $||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$
- $|z_1 - z_2| \leq |z_1| + |z_2|$ (Triangle inequality)
- $||z_1| - |z_2|| \leq |z_1 - z_2|$

These inequalities follow from the **triangle inequality** and the geometric interpretation in the complex plane. The distance between two points cannot be greater than the sum of their distances from the origin, and it is at least the **absolute difference** of those distances.

★ PRODUCT OF MODULI



For any two complex numbers z_1 and z_2 ,

$$|z_1 z_2| = |z_1| |z_2|$$

PROOF

Let z_1 and z_2 be any two complex numbers.

Using $|z|^2 = z \bar{z}$,

$$\begin{aligned} |z_1 z_2|^2 &= (z_1 z_2) \overline{(z_1 z_2)} \\ &= (z_1 z_2) (\bar{z}_1 \bar{z}_2) \\ &= (z_1 \bar{z}_1) (z_2 \bar{z}_2) \\ &= |z_1|^2 |z_2|^2 \end{aligned}$$

Taking square roots on both sides,

$$|z_1 z_2| = |z_1| |z_2|$$

EXTENSION

By mathematical induction, this property can be generalized to any finite number of complex numbers:

$$|z_1 z_2 z_3 \cdots z_n| = |z_1| |z_2| |z_3| \cdots |z_n|$$

That is, the modulus of a product of complex numbers is equal to the product of the moduli of those complex numbers.

QUOTIENT OF MODULI

For any two complex numbers z_1 and z_2 with $z_2 \neq 0$,

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

PROOF

Let z_1 and z_2 be any two complex numbers with $z_2 \neq 0$.

Using $|z|^2 = z\bar{z}$,

$$\begin{aligned} \left| \frac{z_1}{z_2} \right|^2 &= \frac{z_1}{z_2} \cdot \frac{\bar{z}_1}{\bar{z}_2} = \frac{z_1 \bar{z}_1}{z_2 \bar{z}_2} \\ &= \frac{|z_1|^2}{|z_2|^2} \\ \Rightarrow \left| \frac{z_1}{z_2} \right| &= \frac{|z_1|}{|z_2|} \quad (\text{since moduli are non-negative}) \end{aligned}$$

GENERAL FORM

For any non-zero complex numbers $z_1, z_2, \dots, z_n$,

$$\left| \frac{z_1}{z_2 \cdots z_n} \right| = \frac{|z_1|}{|z_2| \cdots |z_n|}$$

GEOMETRIC INTERPRETATION

- Let z_1 and z_2 be represented by vectors $\vec{OZ_1}$ and $\vec{OZ_2}$ in the Argand plane.
- The modulus $|z_1|$ and $|z_2|$ represent the lengths OZ_1 and OZ_2 respectively.
- The complex number $\frac{z_1}{z_2}$ represents the product of z_1 and the reciprocal of z_2 .
- Hence, the length of the vector representing $\frac{z_1}{z_2}$ is the ratio of the lengths OZ_1 and OZ_2 ,
i.e., $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

COMPLEX CONJUGATE

For any complex number $z = a + ib$ (where $a, b \in \mathbb{R}$), the complex conjugate of z is denoted by $\bar{z}$ and defined as

$$\bar{z} = a - ib$$

PROPERTIES

1

$$\overline{\bar{z}} = z$$

The conjugate of the conjugate of z is z itself.

2

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$$

Conjugate of a sum/difference is the sum/difference of the conjugates.

3

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$$

Conjugate of a product is the product of the conjugates.

4

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad z_2 \neq 0$$

Conjugate of a quotient is the quotient of the conjugates.

IMPORTANT RESULTS

- $z + \bar{z} = 2a = 2\text{Re}(z)$

The sum of a complex number and its conjugate is twice its real part.

- $z - \bar{z} = 2ib = 2i\text{Im}(z)$

The difference of a complex number and its conjugate is twice i times its imaginary part.

- $z \bar{z} = |z|^2 = a^2 + b^2$

The product of a complex number and its conjugate is the square of its modulus.

- $|\bar{z}| = |z|$

A complex number and its conjugate have the same modulus.



Thank You